

Technical Report: Dog Adoption Management System

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Contents

1	Introduction	3
1.1	The Evolution of Animal Welfare Systems	3
1.2	Problem Statement	4
1.3	The Proposed Solution	4
1.4	Future Vision and System Evolution	4
1.5	Scope and Significance	4
2	Case Study: Pet Adoption Challenges and Opportunities	5
2.1	The Landscape of Pet Abandonment	5
2.2	The Global Crisis of Pet Overpopulation	5
2.3	The Economic Cost of Abandonment	5
2.4	The Psychological Dimension: Shelter Stress	6
2.5	Data Silos and Information Asymmetry	6
2.6	The Administrative Crisis in Shelters	6
2.7	The Critical Role of Documentation in Responsible Adoption	6
2.8	Detailed Analysis of Current Workflows	7
3	Technical Details of Existing Solutions	8
3.1	Legacy Shelter Management Software	8
3.1.1	Architecture and Technology	8
3.2	Aggregator Marketplaces	8
3.2.1	Architecture and Technology	9
3.3	Generic CRM and Internal Tools	9
3.3.1	Architecture and Technology	9
3.4	Technological Stagnation in the Sector	9
3.5	Comparative Market Matrix	9
4	Overview of Technologies and Motivation	10
4.1	Microservices and Clean Architecture	10
4.2	Backend: .NET 10	10
4.3	Frontend: React	10

4.4	Polyglot Persistence	11
4.5	Orchestration: Docker and AKS	11
4.6	DevOps and CI/CD	11
4.7	Future Technology Stack: AI and Mobile	11
4.8	Monitoring and Observability	11
5	Business Canvas	12
6	Architecture Diagram	13
7	Use Case Diagram	14
8	Functionality, Flow, and API Documentation	15

Chapter 1

Introduction

In the contemporary digital era, the intersection of animal welfare and technology has given rise to innovative solutions aimed at addressing long-standing challenges in the pet adoption process. This technical report details the development and architecture of a sophisticated Dog Adoption Management System, a platform designed to streamline the journey from shelter to forever home.

The primary objective of this project is to create a robust, scalable, and user-centric ecosystem that empowers both animal shelters and potential adopters. By leveraging modern cloud-native technologies, microservices architecture [2], and automated CI/CD pipelines, we aim to provide a seamless experience that reduces the friction inherent in traditional adoption workflows.

Pet adoption is not merely a transactional process; it is a complex emotional and administrative journey. Shelters often struggle with fragmented data, manual record-keeping, and limited reach. Adopters, on the other hand, frequently face opaque application processes, lack of real-time updates, and difficulty in finding the right companion. This project addresses these pain points by centralizing information, automating notifications, and providing an intuitive interface for all stakeholders.

1.1 The Evolution of Animal Welfare Systems

The transition from ledger-based tracking to integrated digital ecosystems represents a paradigm shift in how animal rescues operate. Historically, the focus was purely on "shelter," providing physical safety for animals. In the modern era, the focus has shifted to "rehoming efficiency." The faster an animal can be placed in a permanent home, the more lives a shelter can save. This project is rooted in the belief that software architecture directly impacts animal welfare outcomes.

1.2 Problem Statement

Animal welfare organizations are increasingly overwhelmed by the volume of surrenders and the complexity of modern compliance and tracking requirements. Manual entry systems are prone to human error, resulting in lost medical records or outdated availability listings. Furthermore, the lack of a centralized "source of truth" leads to operational silos, where the marketing team might promote a dog that has already been adopted or is currently under medical quarantine.

1.3 The Proposed Solution

The Dog Adoption Management System proposed in this report utilizes a distributed systems approach. By breaking down the monolith into specialized services, we achieve a higher degree of fault tolerance. The use of Clean Architecture [4] ensures that the business logic remains decoupled from the specific database or cloud provider used, allowing the system to evolve over time without requiring expensive full-scale rewrites.

1.4 Future Vision and System Evolution

While the current implementation focuses on the core orchestration and service delivery, the roadmap for the Dog Adoption Management System includes two major technological leaps. First, the transition from a "3-click" rapid adoption to a rigorous, document-verified workflow ensures that every placement is backed by the necessary legal and residency documentation. Second, we envision a mobile-first expansion utilizing Computer Vision to allow users to identify dog breeds in the real world and instantly connect with available matches in our shelter network. This evolution will transform the platform from a management tool into an intelligent adoption ecosystem.

1.5 Scope and Significance

This report covers the entire lifecycle of the project, from initial case study and market analysis to technical implementation and deployment strategy. The significance of this work lies in its potential to reduce the "Length of Stay" (LOS) for animals in shelters, which is a critical metric in animal welfare. Every day an animal remains in a shelter increases the cost of care and the psychological stress on the animal [3]. By optimizing the administrative pipeline, we directly contribute to better animal outcomes and more efficient shelter operations.

Chapter 2

Case Study: Pet Adoption Challenges and Opportunities

2.1 The Landscape of Pet Abandonment

Pet abandonment remains a global crisis with profound ethical, social, and economic implications. Every year, millions of domestic animals, particularly dogs, end up in shelters or on the streets. In more developed nations, the burden falls primarily on the shelter infrastructure, whereas in developing regions, it poses a significant public health risk due to rabies and other zoonotic diseases [7].

According to global animal welfare organizations, the reasons for abandonment range from socio-economic factors to a lack of understanding of the responsibilities associated with pet ownership. In many cases, behavioral issues, life transitions, and financial constraints drive owners to surrender their companions.

2.2 The Global Crisis of Pet Overpopulation

Pet abandonment is a subset of the larger issue of pet overpopulation. Statistics indicate that in the United States alone, approximately 6.3 million companion animals enter shelters every year [1]. Of these, roughly 3.1 million are dogs. While adoption rates have been increasing, approximately 390,000 shelter dogs are still euthanized annually [1]. This is often not due to health issues, but due to a lack of space and administrative inability to process adoptions fast enough.

2.3 The Economic Cost of Abandonment

The economic impact is often overlooked. Municipalities spend billions of dollars on animal control and shelter operations. A single "intake to adoption" cycle for a dog can

cost anywhere from \$300 to \$800, covering food, medical care, and marketing. When administrative inefficiencies delay an adoption by even a few days, the cumulative cost across hundreds of animals can be substantial.

2.4 The Psychological Dimension: Shelter Stress

Modern veterinary science has documented "shelter stress." Dogs in a shelter environment often exhibit chronic anxiety and suppressed immune systems [3]. A digital system that fast-tracks the application and matching process directly contributes to the mental well-being of the animals by reducing their "length of stay" (LOS).

2.5 Data Silos and Information Asymmetry

One of the most persistent bottlenecks identified in current shelter operations is "Information Asymmetry." Information known to the intake officer (e.g., a dog's fear of thunderstorms) often doesn't reach the marketing volunteer or the final adopter until after a placement failure occurs.

By implementing a document-based NoSQL storage for pet profiles, our system allows for the storage of unstructured, rich metadata that can be updated by different departments simultaneously. This breaks the "linear" flow of information and creates a collaborative data environment.

2.6 The Administrative Crisis in Shelters

Most animal shelters operate on very tight budgets. Staff and volunteers must manage complex intake records, medical histories, and hundreds of adoption applications. The reliance on manual or fragmented systems leads to massive inefficiencies. For example, a dog might be listed as "Available" long after an application has been approved, leading to frustration for potential adopters and redundant work for staff.

2.7 The Critical Role of Documentation in Responsible Adoption

A major challenge identified during our research is the balance between a "seamless" digital experience and the need for due diligence. While rapid adoption can reduce shelter load, a "convoluted" or rather *rigorous* documentation process is often necessary to ensure the long-term safety of the animal. This involves the verification of housing agreements, identification, and previous pet history. Future iterations of this project will integrate an

automated document verification microservice to handle these complex legal requirements without increasing the manual workload of shelter staff.

2.8 Detailed Analysis of Current Workflows

In a typical non-digitized shelter, an adoption application involves a multi-step paper trail. The applicant fills out a form, which is then manually reviewed by a volunteer. This review often requires checking a physical folder for the dog's latest medical status. If the volunteer is part-time, the folder might not be accessed for 48 hours. By the time the review is complete, the dog's status may have changed. Our project aims to replace this asynchronous, error-prone workflow with a real-time, event-driven architecture that ensures all stakeholders have access to the same, current data at all times.

Chapter 3

Technical Details of Existing Solutions

To understand the value proposition of the Dog Adoption Management System, it is crucial to analyze the landscape of currently available tools. Existing solutions generally fall into three categories: Legacy Shelter Management Software, Aggregator Marketplaces, and Generic CRM/Internal tools.

3.1 Legacy Shelter Management Software

These are specialized applications designed specifically for shelters. Examples include "ShelterBuddy" and "PetPoint".

3.1.1 Architecture and Technology

Historically, these systems were built as monolithic applications. Many still rely on early cloud models that lack true elasticity. Their architectures often suffer from:

- **High Coupling:** This architectural debt makes it nearly impossible to implement rapid feature updates. A change in the intake logic might inadvertently break the adoption contract generation.
- **Data Inaccessibility:** Many legacy providers lock shelters into their ecosystem by making API access expensive or non-existent.
- **Monolithic Scaling:** If one part of the app (like image processing) is under heavy load, the entire system slows down.

3.2 Aggregator Marketplaces

Platforms like Petfinder and Adopt-a-Pet are the most visible solutions.

3.2.1 Architecture and Technology

These are modern, web-scale platforms optimized for search. They use sophisticated indexing and CDNs. However, they are primarily marketing tools, not management tools. Shelters still need a separate system to handle internal administrative data. This creates a synchronization problem where a shelter must manually update marketplaces every time a dog's status changes.

3.3 Generic CRM and Internal Tools

Some organizations use general-purpose tools like Salesforce or Trello.

3.3.1 Architecture and Technology

These tools offer flexibility through multi-tenant SaaS architectures. However, they require a "custom developer" to map pet welfare requirements to generic fields. For example, a "Lead" becomes an "Adopter." This often results in a "clunky" user experience that doesn't feel native to the mission.

3.4 Technological Stagnation in the Sector

The primary issue across all existing categories is technological stagnation. While the broader tech industry has moved toward serverless, microservices, and container orchestration [2], the animal welfare sector remains stuck with platforms that are difficult to scale and integrate. The Dog Adoption Management System leverages AKS and Docker to ensure that the infrastructure is as modern and agile as the code it runs.

3.5 Comparative Market Matrix

Feature	Legacy SW	Marketplaces	Generic CRM	Our Project
API-First	No	Partial	Yes	Yes
Clean Arch	No	No	N/A	Yes
Scalability	Low	High	High	High
Specialized Logic	High	Low	Low	High

Table 3.1: Comparison of adoption management solutions

Chapter 4

Overview of Technologies and Motivation

The Dog Adoption Management System is built using a modern, cloud-native stack designed for high availability and scalability. The selection of these technologies was driven by the need to handle complex business logic while maintaining a professional user experience.

4.1 Microservices and Clean Architecture

The project adopts a **Microservices Architecture**, which allows each functional area to be developed and scaled independently [2]. Within each service, we implement **Clean Architecture** [4]. This ensures that the core business logic is decoupled from external technical concerns like databases or web frameworks.

4.2 Backend: .NET 10

The backend is built using **.NET 10** [5]. .NET was chosen for its high performance, robust NuGet ecosystem, and first-class support for Azure integrations. We use **Entity Framework Core** for ORM and **MediatR** to implement the CQRS pattern.

4.3 Frontend: React

The UI is a Single Page Application (SPA) built with **React** and **Vite**. We use **Tailwind CSS** for a utility-first approach to styling, ensuring the platform is responsive and consistent.

4.4 Polyglot Persistence

We utilize a polyglot persistence strategy:

- **PostgreSQL:** For structured, relational data like user profiles and reviews.
- **Azure CosmosDB:** For highly scalable, document-based adoption applications [6].

4.5 Orchestration: Docker and AKS

All components are containerized using **Docker**. The system is orchestrated by **Azure Kubernetes Service (AKS)**, which provides self-healing, auto-scaling, and a managed control plane for the microservices.

4.6 DevOps and CI/CD

We use **GitHub Actions** for automated CI/CD. The pipeline is secured using **OIDC (OpenID Connect)** for password-less authentication between GitHub and Azure. On every push to the main branch, all 8 container images are built, pushed to the **Azure Container Registry**, and deployed to the cluster.

4.7 Future Technology Stack: AI and Mobile

To realize our vision for a breed-scanning mobile feature, we plan to incorporate:

- **Computer Vision (TensorFlow/PyTorch):** For implementing Convolutional Neural Networks (CNNs) that can classify dog breeds from mobile camera streams.
- **Mobile Development (React Native / Flutter):** To extend our web-based ecosystem into a cross-platform mobile application.
- **Identity and Document Verification (Azure AI Vision):** To automate the "convoluted" document verification process, ensuring that submitted IDs and housing permits are authentic and valid.

4.8 Monitoring and Observability

Azure Application Insights is integrated into the core services to provide real-time telemetry. This allows the team to monitor "Success Rates" of applications and identify performance bottlenecks in the distributed system.

Chapter 5

Business Canvas

Chapter 6

Architecture Diagram

Chapter 7

Use Case Diagram

Chapter 8

Functionality, Flow, and API Documentation

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